Exp 17 : PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE DATE: WHITE GAUSSIAN NOISE (AWGN) USING MATLAB

Objectives :

1. To generate a digital communication signal in MATLAB and transmit it through an Additive White Gaussian Noise (AWGN) channel.
2. To obtain and analyze the eye diagram of the received digital signal for different noise conditions.
3. To measure eye-opening parameters such as eye height, eye width, crossing level and timing variation.
4. To study the effect of decreasing SNR / Eb/N0 on inter-symbol interference, noise margin and timing performance.
5. To compare the transmitted and received signal characteristics and evaluate communication-system performance.

Code :

```
clc;
clear;
close all;

%% Parameters
N = 100; % Number of bits
sps = 20; % Samples per symbol
SNR = 10; % Signal-to-Noise Ratio (dB)

%% Generate Random Binary Data
data = randi([0 1],1,N);

%% NRZ Encoding (0 -> 0, 1 -> 1)
tx_signal = repelem(data,sps);

%% Pass Signal through AWGN Channel
rx_signal = awgn(tx_signal,SNR,'measured');

%% Time Axis
t = (0:length(tx_signal)-1)/sps;
```

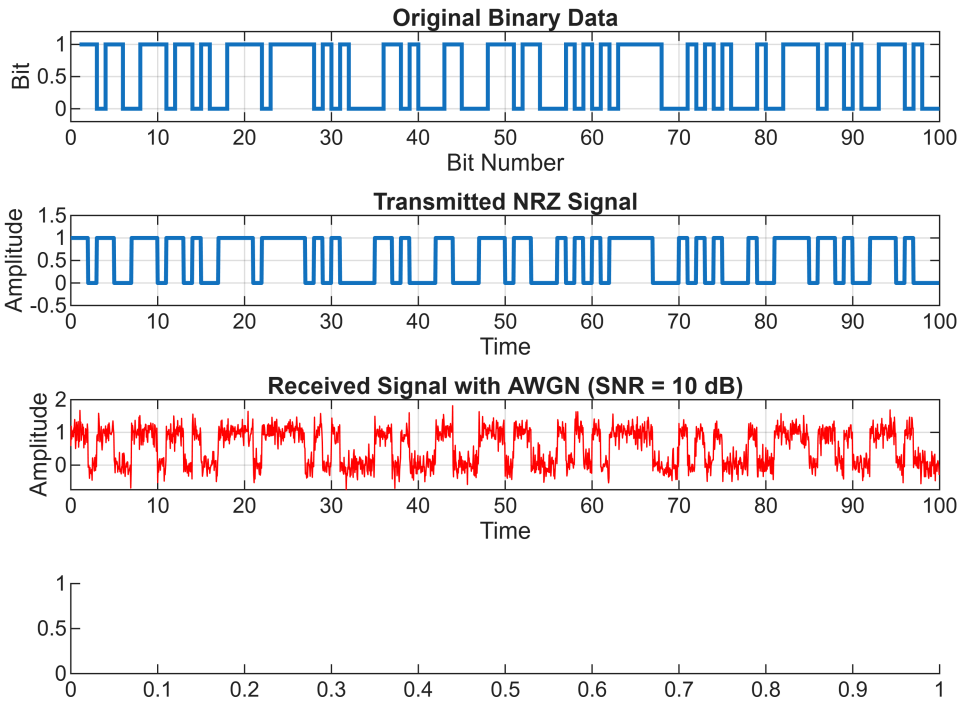
```
%% Plot Results
figure;

% Original Binary Data
subplot(4,1,1)
stairs(data,'LineWidth',1.5)
grid on
title('Original Binary Data')
xlabel('Bit Number')
ylabel('Bit')
ylim([-0.2 1.2])

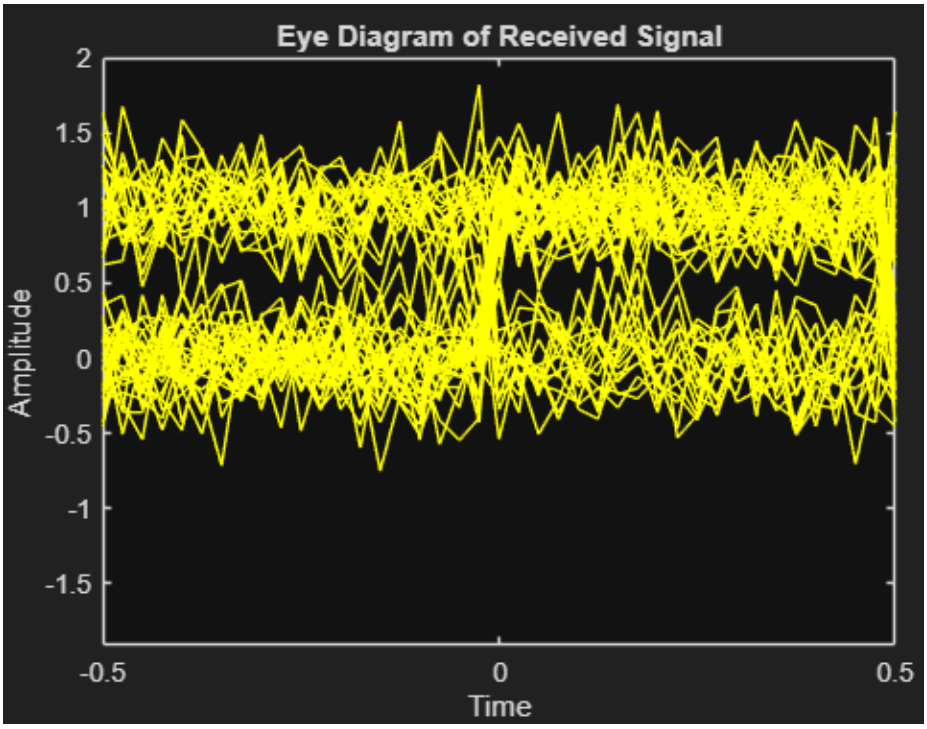
% Transmitted NRZ Signal
subplot(4,1,2)
plot(t,tx_signal,'LineWidth',1.5)
grid on
title('Transmitted NRZ Signal')
xlabel('Time')
ylabel('Amplitude')
ylim([-0.5 1.5])

% Received Signal after AWGN
subplot(4,1,3)
plot(t,rx_signal,'r')
grid on
title(['Received Signal with AWGN (SNR = ',num2str(SNR),' dB)'])
xlabel('Time')
ylabel('Amplitude')

% Eye Diagram
subplot(4,1,4)
```




```
eyediagram(rx_signal,2*sps)
title('Eye Diagram of Received Signal')
```



Result :

i) Successfully generated an NRZ digital communication signal and transmitted it through an AWGN channel.

ii) Observed that the received signal was affected by Gaussian noise, resulting in signal distortion and bit errors.

iii) Successfully plotted the eye diagram of the received signal using MATLAB.

iv) Observed that the eye opening decreased due to AWGN, indicating reduced signal quality and increased probability of bit errors.